

RAGtune Architecture

RAGtune is a budget-aware middleware designed to sit between your application and various retrieval/reranking backends. It treats **cost** and **latency** as first-class constraints.



RAGtune Architecture Diagram

Core Philosophy: Active Learning & Feedback Loops

Unlike traditional RAG pipelines that follow a static "waterfall" (Retrieve -> Rerank -> Final), RAGtune implements an **Active Learning** approach using an iterative feedback loop:

1. **Sample**: The system picks a small batch of documents.
2. **Measure**: Ground-truth scores are obtained from expensive rerankers or LLMs.
3. **Learn**: The **Utility Estimator** updates the predicted utility of the remaining *unranked* documents based on these results.